Reg. No. [ ] [18AU51]

FIFTH SEMESTER B.Tech EXAMINATION,DECEMBER-2021

VEHICLE DYNAMICS

Time: 3 Hours **Maximum Marks: 70**

Instructions:

i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.

ii. Missing data may be suitably assumed.

1.	a.	Define the free vibration and brief the types of free vibrations with neat diagram.	08
	b.	Briefly Explain the single degree and Two degree freedom systems with respective equations of motion.	06
2.	a.	Define the term damping and explain in detail the types of damping.	06
	b.	Calculate the natural frequency of vibration of a torsional pendulum with following dimensions. Length of rod l=1m Diameter of rod d=5mm Diameter of rotor D=0.2m Mass of rotor M= 2kg Modulus of rigidity for the material of rod assumed to be $0.83*10^{11}$ N/m^2	08
3.	a.	Explain the Calculation of acceleration in detail.	09
	b.	Briefly Explain the overturning of a vehicle.	05
		OR	
4.	a.	What are the functions of Tractive effort?	06
	b.	Briefly explain the Lateral stability of vehicles.	08
5.	a.	Discuss in detail the wheel hop. Wheel wobble, sprung mass frequency and the function of suspension.	10
	b.	What are the requirements, elements of suspension?	04
		OR	
6.	a.	Discuss in detail the common causes of steering wheel shakes when braking.	08
	b.	Brief about the roll axis of an automobile.	06
7.	a.	Define tires and brief in detail the manufacturing process of tires.	07
	b.	Briefly explain the ride characteristics of tires.	07
		OR	
8.	a.	Briefly explain the power losses in tires.	08
	b.	Discuss in detail the over steer and under steer of an automobile.	06